

NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

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SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 8: L12-L13

Question 1 (Lecture 12)

“If a car won’t start and battery is old, then replacing the battery might fix the issue”. What type of knowledge does this statement belong to?

- (a) Declarative
- (b) Procedural
- (c) Structural
- (d) Heuristic
- (e) Meta

Answer: (d). Experience-based guideline: replacing the battery might fix the issue. The solution is not guaranteed to be correct/optimal.

Question 2 (Lecture 12)

Which of the following statements represent meta knowledge?

- (a) “The Eiffel Tower is in Paris.” (Declarative)
- (b) “The steps to solve a quadratic equation are: 1) rearrange terms, 2) apply quadratic formula.” (Procedural)
- (c) “If the reasoning process takes too long, switch from model checking to resolution.”
- (d) “The ground is wet. It possibly rained last night.” (Heuristic)

Answer: (c). It’s about choosing a reasoning strategy.

Question 3 (Lecture 12)

In which scenario is probabilistic knowledge representation most useful?

- (a) When all facts are certain and must be represented with strict rules. (Logical)
- (b) When handling noisy sensor data with uncertain outcomes.
- (c) When representing fixed hierarchical relationships, like a taxonomy of animals. (Structured)
- (d) When encoding concepts as high-dimensional vectors for natural language processing. (Distributed)

Answer: (b)

Question 4 (Lecture 12)

What type of reasoning does propositional logic use?

- (a) Deductive
- (b) Inductive

(c) Abductive

Answer: (a). Deterministic in nature.

Question 5 (Lecture 13)

A single proposition symbol can represent a compound proposition.

- (a) True
- (b) False

Answer: (b). A proposition symbol can only represent an atomic proposition.

Question 6 (Lecture 13)

Which of the following is/are a well-formed formula in propositional logic?

- (a) $\alpha = \gamma \Leftrightarrow \delta$
- (b) $\neg\neg\alpha$, provided that α is a well-formed formula
- (c) $\neg\alpha \vee \beta \Rightarrow \beta$, provided that α and β are both well-formed formulas.
- (d) $\alpha \wedge \sigma \wedge \neg\beta$
- (e) $\beta \models \alpha$, provided that α and β are both well-formed formulas.

Answer: (b)(c)

Question 7 (Lecture 13)

Which of the following is/are correct?

- (a) $(A \wedge B) \models (A \vee B)$
- (b) $\alpha \models \beta$ if and only if $(\alpha \Rightarrow \beta)$ is a tautology
- (c) $\alpha \equiv \beta$ if and only if $(\alpha \Leftrightarrow \beta)$ is a tautology
- (d) $(A \wedge B) \wedge (A \Leftrightarrow B)$ is satisfiable

Answer: (a)(b)(c)(d)

(a) Since “if $(A \wedge B)$ is true, $(A \vee B)$ will always be true”, $(A \wedge B) \models (A \vee B)$

(b) Need to check: ① if $\alpha \models \beta$, $(\alpha \Rightarrow \beta)$ is always true; and ② if $(\alpha \Rightarrow \beta)$ is always true, $\alpha \models \beta$.

① If $\alpha \models \beta$, that said, whenever α is true, β is true. If α is true, β is true. Then $(\alpha \Rightarrow \beta)$ is true; if α is false, no matter what truth-value β takes, $(\alpha \Rightarrow \beta)$ is true. Therefore, $(\alpha \Rightarrow \beta)$ is always true if $\alpha \models \beta$.

② if $(\alpha \Rightarrow \beta)$ is always true, that said, only 3 pairs of truth value assignments

for (α, β) are allowed: (true, true), (false, true), (false, false). In these 3 models, β is always true if α is true, which means $\alpha \models \beta$.

(c) Similar to (b). Need to check: ① if $\alpha \models \beta$, $(\alpha \Leftrightarrow \beta)$ is always true; and ② if $(\alpha \Leftrightarrow \beta)$ is always true, $\alpha \models \beta$.

(d) Need to check that $(A \wedge B) \wedge (A \Leftrightarrow B)$ is true in at least one model.

$(A \wedge B)$ and $(A \Leftrightarrow B)$ should both be true if we want $(A \wedge B) \wedge (A \Leftrightarrow B)$ to be true. Then, A and B should both be true if we want $(A \wedge B)$ to be true; A and B should both be true or false if we want $(A \Leftrightarrow B)$ to be true. Then, we find out one model which is $\{A=\text{true}, B=\text{true}\}$ that satisfies it.

Question 8 (Lecture 13)

On an island, Knights always tell the truth and Knaves always lie.

Person A says: “Person B is a Knave.”

Person B says: “We are both Knights.”

(i) Construct the knowledge base using propositional logic. Hint: you may use “sympy” library.

(ii) Determine who is Knight and who is Knave.

Answer: See TUT8_Q8.ipynb

(i) Proposition Symbols: A: “Person A is a Knight”; B: “Person B is a Knight”

$$KB = \{A \Leftrightarrow \neg B, B \Leftrightarrow (A \wedge B)\}$$

(ii) Check if $KB \models A$, $KB \models B$. Since $KB \models A$, A is a Knight. Since KB does not entail B, we further check if $KB \models \neg B$. Since $KB \models \neg B$ (i.e., B is always false whenever KB is true), B is a Knave.

Question 9 (Lecture 13)

Consider a simple 3x3 Wumpus World, where at least one Wumpus and at least one pit are located in some rooms. In the rooms directly (not diagonally) adjacent to the Wumpus, the agent will perceive a Stench. In the squares directly adjacent to a pit, the agent will perceive a Breeze.

(i) Construct a simple knowledge base using propositional logic for this 3x3 Wumpus World. You may use “sympy” library. Use the following proposition symbol for each $[x,y]$ location:

$P_{x,y}$: There is a pit in $[x,y]$

$W_{x,y}$: There is a Wumpus in $[x,y]$

$B_{x,y}$: The agent perceives a breeze in $[x,y]$

$S_{x,y}$: *The agent perceives a stench in $[x, y]$*

- (ii) Suppose the agent is now in location $[1,1]$, where neither Breeze nor Stench is perceived. Assume there is neither pit nor Wumpus in location $[1,1]$. Is there a Wumpus in location $[2, 2]$? Is there a pit in location $[2, 2]$?
- (iii) Continuing from part (ii), are adjacent cells of location $[1,1]$ safe? Suppose the agent moves to location $[2,1]$ and perceive a Breeze but no Stench. Are adjacent cells of location $[2,1]$ safe?

Answer: See [TUT8_Q9.ipynb](#)

(ii) Not sure.

(iii) Adjacent cells of location $[1,1]$ are safe;

Adjacent cells of location $[2,1]$: $[1,1]$ safe; $[3,1]$ no Wumpus, not sure about pit; $[2,2]$ no Wumpus, not sure about pit.